Homework 2.

Problem 1

Verify kraft. $\sum_{i}^{N} 2^{-l_i} < 1$

for $\{2\ 2\ 3\ 3\ 3\ 3\}$.

$\sum_{i}^{N} 2^{-l_i} = 2^{-2}+2^{-2}+2^{-3}+2^{-3}+2^{-3} = 7/8 \le 1$

2 11
2 10
3 010
3 001
3 000

Problem 2:

uniquely decodeable condition is

if $x_i \neq x_j$, $C(x_i) \neq C(x_j)$

在不使用前缀码的约束下. 不妨考虑后缀码.

00, 001, 10, 101, 011

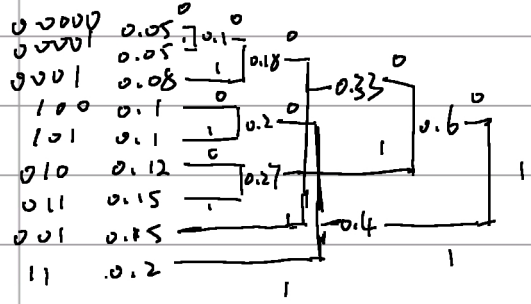
不是前缀, 但任何码字都不是其他码字的后缀.

仅向解码时, 其实为前缀码, 而前缀码是uniquely decodeable的. 因此, 后缀码亦是uniquely decodeable的.

Problem 3

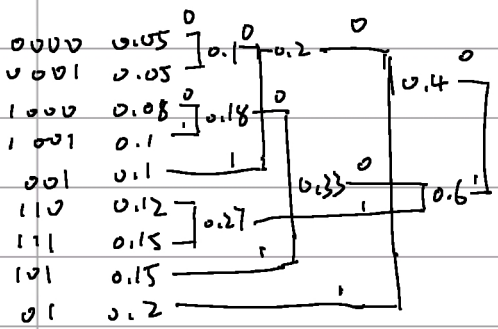
For $\{0.1\ \ 0.12\ \ 0.08\ \ 0.15\ \ 0.05\ \ 0.05\ \ 0.2\ \ 0.1\ \ 0.15\} = X$

$H(X) = -\sum_{i}^{9} p_i \log_2 p_i \approx 3.04$ bits/symbol



平均码长

$(0.05+0.05)\times 5 + (0.1+0.1+0.12+0.15+0.08\times 4) + (0.15)\times 3 + 0.2\times 2 = 3.08$ bits.



平均码长 3.08 bits.

Problem 4

2 2 2 3 4 4

A B C D E F

D在上一层.

$p_E \le p_F,\ p_D \le p_F \Rightarrow p_E + p_F \le p_A / p_B / p_C$

又 $\because p_E \ge 0 \Rightarrow p_F \le p_A, p_B, p_C$

$\therefore p_A + p_B + p_C + p_D + p_F \ge 5p_F$

$1 = p_A + p_B + p_C + p_D + p_E + p_F \Rightarrow p_F \le 1/5$